Power Series and Interval of Convergence

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Finding the radius of convergence**

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ using the Ratio Test.

$\lim_{n\to\infty} \left|\frac{x^{n+1}/(n+1)}{x^n/n}\right| = \lim_{n\to\infty} \frac{n|x|}{n+1} = |x|$. Converges when $|x| < 1$: radius of convergence $R = 1$.

2 **[Foundational]** • **Target: Finding the interval of convergence (testing endpoints)**

Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ (radius $R = 1$ found earlier), by testing $x = 1$ and $x = -1$.

At $x = 1$: $\sum \frac{1}{n}$ diverges (harmonic). At $x = -1$: $\sum \frac{(-1)^n}{n}$ converges (alternating harmonic, by AST). Interval of convergence: $[-1, 1)$.

3 **[Foundational]** • **Target: Representing functions as power series**

Write the power series for $\frac{1}{1-x}$ using the geometric series formula, and state its interval of convergence.

$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$, valid for $|x| < 1$, i.e. interval of convergence $(-1, 1)$.

4 **[Foundational]** • **Target: Representing functions as power series**

Use the geometric series for $\frac{1}{1-x}$ to find a power series for $\frac{1}{1+x^2}$.

Substitute $x \to -x^2$: $\frac{1}{1-(-x^2)} = \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n} = 1 - x^2 + x^4 - x^6 + \cdots$.

5 **[Foundational]** • **Target: Differentiating and integrating power series**

Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, find a power series for $\frac{1}{(1-x)^2}$ by differentiating term by term.

$\frac{d}{dx}\left[\frac{1}{1-x}\right] = \frac{1}{(1-x)^2}$. Differentiating the series: $\sum_{n=1}^{\infty} n x^{n-1}$. So $\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1} = 1 + 2x + 3x^2 + \cdots$.

6 **[Foundational]** • **Target: Differentiating and integrating power series**

Using $\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$, find a power series for $\tan^{-1} x$ by integrating term by term.

$\tan^{-1} x = \int \frac{1}{1+x^2}\,dx = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} + C$. Since $\tan^{-1} 0 = 0$, $C = 0$: $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$.

7 **[Foundational]** • **Target: More radius/interval of convergence practice**

Find the radius and interval of convergence of $\sum_{n=0}^{\infty} n!\, x^n$.

$\lim_{n\to\infty} \left|\frac{(n+1)!\, x^{n+1}}{n!\, x^n}\right| = \lim_{n\to\infty} (n+1)|x| = \infty$ for any $x \neq 0$. Converges only at $x = 0$: $R = 0$, interval $\{0\}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Visualizing a power series approximation

The geometric series $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$ converges on $(-1, 1)$. Compare the graph of $f(x) = \frac{1}{1-x}$ to its degree-3 partial sum $P_3(x) = 1 + x + x^2 + x^3$ on $[-0.8, 0.8]$, and describe where they agree most closely.

The two curves nearly overlap near $x = 0$ and diverge visibly as $x \to \pm 0.8$ (closer to the boundary of the interval of convergence), since the partial sum's approximation quality worsens near the edges of convergence.

9 [Intermediate] • Target: Term-by-term operations preserve the radius of convergence

State the theorem describing what happens to the radius of convergence when a power series is differentiated or integrated term by term.

Differentiating or integrating a power series term by term does not change its radius of convergence R (though convergence AT the endpoints $x = \pm R$ may change).

10 [Intermediate] • Target: Term-by-term operations preserve the radius of convergence

Given that $\sum_{n=1}^{\infty} \frac{x^n}{n}$ has radius of convergence $R = 1$, state the radius of convergence of its derivative $\sum_{n=1}^{\infty} x^{n-1}$ without recomputing.

By the term-by-term differentiation theorem, the radius of convergence is unchanged: $R = 1$.

11 [Intermediate] • Target: Building a new series by combining known ones

Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ and $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$, find the power series for $\frac{2}{1-x^2}$ by adding the two series.

$\frac{1}{1-x} + \frac{1}{1+x} = \frac{(1+x)+(1-x)}{1-x^2} = \frac{2}{1-x^2}$. Adding the series: $\sum_{n=0}^{\infty} x^n + \sum_{n=0}^{\infty} (-1)^n x^n = \sum_{n=0}^{\infty} [1 + (-1)^n] x^n$, which is $2x^n$ for even n and 0 for odd n: $2 + 2x^2 + 2x^4 + \cdots$.

12 [Intermediate] • Target: Ratio Test with factorial and power combinations

For the series above, test both endpoints $x = \frac{1}{2}$ and $x = -\frac{1}{2}$ to find the full interval of convergence.

At $x = \frac{1}{2}$: $\sum \frac{2^n (1/2)^n}{n^2} = \sum \frac{1}{n^2}$, converges (p-series, $p = 2$). At $x = -\frac{1}{2}$: $\sum \frac{(-1)^n}{n^2}$, converges absolutely. Interval: $\left[-\frac{1}{2}, \frac{1}{2}\right]$.

13 [Intermediate] • Target: Using a power series to approximate a definite integral

Using the power series $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$ found earlier, estimate $\tan^{-1}(0.5)$ using the first three terms.

$\tan^{-1}(0.5) \approx 0.5 - \frac{(0.5)^3}{3} + \frac{(0.5)^5}{5} = 0.5 - 0.04167 + 0.00625 \approx 0.4646$ (true value ≈ 0.4636, a close match).

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14 **[Intermediate]** • **Target: Differentiating and integrating power series**

Using the binomial series for $(1-x^2)^{-1/2}$, derive the Maclaurin series expansion for $f(x)=\arcsin x$ up to and including the x^5 term. State its radius of convergence.

1. Recall the general binomial expansion $(1+u)^p = 1+pu+\frac{p(p-1)}{2!}u^2+\ldots$. Here $u=-x^2$ and $p=-1/2$: $(1-x^2)^{-1/2} = 1+\left(-\frac{1}{2}\right)(-x^2)+\frac{(-1/2)(-3/2)}{2}(-x^2)^2+O(x^6) = 1+\frac{1}{2}x^2+\frac{3}{8}x^4+O(x^6)$. 2. Since $\arcsin x = \int_0^x \frac{1}{\sqrt{1-t^2}}\,dt$, integrate term-by-term: $\arcsin x = \int_0^x \left(1+\frac{1}{2}t^2+\frac{3}{8}t^4+\ldots\right)dt = x+\frac{1}{6}x^3+\frac{3}{40}x^5+\ldots$ 3. The binomial series converge.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: More radius/interval of convergence practice** • **Cross-Lesson: 3.01 Parametric Equations**

Find the Maclaurin series for $\ln(1+2x)$ up to the x^3 term, and state its interval of convergence.

Use $\ln(1+u) = u-\frac{u^2}{2}+\frac{u^3}{3}-\ldots$ with $u=2x$: $\ln(1+2x) = 2x-2x^2+\frac{8}{3}x^3-\ldots$. Convergence requires $-1<2x\le 1 \Rightarrow -\frac{1}{2}<x\le\frac{1}{2}$.

16 **[Challenging]** • **Target: Visualizing a power series approximation** • **Cross-Lesson: 3.02 Vector-Valued Functions**

Level 3 · AdvancedEvaluate the definite integral using a substitution that transforms the limits: $\int_0^{\sqrt{\pi/2}} x\cos(x^2)e^{\sin(x^2)}\,dx$

Let $u=\sin(x^2)$. Differentiating gives $du=\cos(x^2)\cdot 2x\,dx = 2x\cos(x^2)\,dx \Rightarrow x\cos(x^2)\,dx=\frac{1}{2}du$. Transform the integration limits:Lower limit $x=0\Rightarrow u=\sin(0)=0$.Upper limit $x=\sqrt{\pi/2}\Rightarrow u=\sin((\sqrt{\pi/2})^2)=\sin(\pi/2)=1$.Substitute into the integral.

17 **[Challenging]** • **Target: Term-by-term operations preserve the radius of convergence** • **Cross-Lesson: 4.01 Polar Curves and Derivatives**

Level 1 · FoundationsExpress the infinite summation limit as a definite integral on the interval $[0,1]$ and evaluate it: $\lim_{n\to\infty}\sum_{i=1}^{n}\frac{i^3}{n^4}$.

Factor $\frac{1}{n}=\Delta x$: $\sum_{i=1}^{n}\frac{i^3}{n^4} = \sum^{n}\left(\frac{i}{n}\right)^3\cdot\frac{1}{n}$. Here $x_i=\frac{i}{n}=0+i\Delta x$ with $\Delta x=\frac{1}{n}$, matching the right Riemann sum of $f(x)=x^3$ on $[0,1]$: $\lim_{n\to\infty}\sum_{i=1}\left(\frac{i}{n}\right)^3\frac{1}{n}=\int_0^1 x^3\,dx=\left[\frac{x^4}{4}\right]_0^1=\frac{1^4}{4}-0=\mathbf{\frac{1}{4}}$.

18 **[Challenging]** • **Target: Building a new series by combining known ones** • **Cross-Lesson: 4.02 Area in Polar Coordinates**

Use the substitution $x=2\sin\theta$ to evaluate the exact definite integral $\int_0^1\sqrt{4-x^2}\,dx$.

1. Substitute $x=2\sin\theta\Rightarrow dx=2\cos\theta\,d\theta$. When $x=0$, $\theta=0$. When $x=1$, $2\sin\theta=1\Rightarrow\sin\theta=\frac{1}{2}\Rightarrow\theta=\frac{\pi}{6}$. 2. Rewrite integrand: $\sqrt{4-x^2}=\sqrt{4-4\sin^2\theta}=2\sqrt{\cos^2\theta}=2\cos\theta$ (since $\cos\theta>0$ on $[0,\pi/6]$). 3. Substitute into the integral.

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19 [Challenging] • Target: Ratio Test with factorial and power combinations • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Level 3 · AdvancedEvaluate the definite integral analytically: $\int_{-3}^{3}\left(x^5\sqrt{9-x^2}+\sqrt{9-x^2}\right)dx$.

Split the integral using linearity: $\int_{-3}^{3}x^5\sqrt{9-x^2}\,dx+\int_{-3}^{3}\sqrt{9-x^2}\,dx$. Part 1 · Symmetry Analysis: Let $g(x)=x^5\sqrt{9-x^2}$. Then $g(-x)=(-x)^5\sqrt{9-(-x)^2}=-x^5\sqrt{9-x^2}=-g(x)$. Because g is an odd function integrated over symmetric limits $[-3,3]$: $\int_{-3}^{3}x^5\sqrt{9-x^2}\,dx=0$. Part 2 · Geometric Area: The integrand $y=\sqrt{9-x^2}\geq 0$ represents the upper half of the circle $x^2+y^2=9$ with radius $r=3$. The integral from.

20 [Challenging] • Target: Using a power series to approximate a definite integral • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Level 2 · Medium Evaluate the indefinite integral using partial fraction decomposition: $\int\frac{5x-3}{x^2-2x-3}\,dx$

Factor the denominator: $x^2-2x-3=(x-3)(x+1)$. Set up partial fractions: $\frac{5x-3}{(x-3)(x+1)}=\frac{A}{x-3}+\frac{B}{x+1}\Rightarrow 5x-3=A(x+1)+B(x-3)$ Using Heaviside cover-up: Set $x=3$: $5(3)-3=A(3+1)\Rightarrow 12=4A\Rightarrow A=3$. Set $x=-1$: $5(-1)-3=B(-1-3)\Rightarrow -8=-4B\Rightarrow B=2$. Integrate term-by-term.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Finding the radius of convergence

Find the full interval of convergence of $\sum_{n=1}^{\infty}\frac{(x+7)^n}{3^n}$.

a. [-10,-4]

b. [-10,-4)

c. **(-10,-4)** ✓

d. (-10,-4]

Ratio Test gives $|x+7|<3$, so $-10<x<-4$ ($R=3$). Testing endpoints: the terms don't approach 0 at either endpoint, so both diverge (by the nth-term test). Interval: $(-10,\ -4)$.

22 [Foundational] • Target: Finding the interval of convergence (testing endpoints)

Using $\frac{1}{1-3x}=\sum_{n=0}^{\infty}3^nx^n$, find the coefficient of x^3.

a. 27 ✓

b. 26

c. 81

d. 9

The coefficient of x^n is 3^n, so the coefficient of x^3 is $3^3=27$.

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23 [Foundational] • Target: Representing functions as power series

Integrating $\frac{1}{1+x^2} = \sum_{k=0}^{\infty} (-1)^k x^{2k}$ term-by-term gives the Maclaurin series for $\arctan x$. Find the coefficient of x^{21}.

a. $\frac{2}{21}$

b. $-\frac{1}{21}$

c. $\frac{1}{21}$ ✓

d. $\frac{1}{22}$

$\int \sum (-1)^k x^{2k} dx = \sum \frac{(-1)^k x^{2k+1}}{2k+1}$. At $2k+1 = 21$: coefficient $= \frac{(-1)^{10}}{21} = \frac{1}{21}$.

24 [Foundational] • Target: Differentiating and integrating power series

Differentiating the Maclaurin series $\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$ term-by-term gives the series for $\frac{1}{(1-x)^2}$. Find the coefficient of x^3.

a. 6

b. 8

c. **4** ✓

d. 3

$\frac{d}{dx} \sum x^k = \sum k x^{k-1} = \frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} (n+1) x^n$. Coefficient of x^3 is $3+1=4$.

25 [Foundational] • Target: More radius/interval of convergence practice

Interval of convergence of $\sum \frac{(-3)^n (x-2)^n}{\sqrt{n}}$: centre, radius, and endpoints:

a. $R = 1/3$, **interval** $(5/3, 7/3]$ ✓

b. $R = 1/3$, interval $[5/3, 7/3)$

c. $R = 1/3$, interval $[5/3, 7/3]$

d. $R = 3$, interval $(-1, 5]$

$|-3(x-2)| < 1 \Rightarrow |x-2| < 1/3$. At $x = 7/3$: $\sum (-1)^n/\sqrt{n}$ converges (Leibniz). At $x = 5/3$: $\sum 1/\sqrt{n}$ diverges.

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26 [Foundational] • Target: Visualizing a power series approximation

The function $f(x)$ is represented by the power series $f(x) = \sum_{n=0}^{\infty} a_n(x-c)^n$. It is known that $f(0) = 2$, $f'(0) = 3$, and $f''(0) = 12$. If the Taylor series expansion of $f(x)$ around $x = 0$ is given by $f(x) = 2 + 3x + bx^2 + \ldots$, find the value of the coefficient b and the center c.

a. $b = 6, c = 0$ ✓

b. $b = 6, c = 1$

c. $b = 12, c = 0$

d. $b = 2, c = 0$

The Taylor series expansion of a function $f(x)$ around $x = c$ is given by $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n$. In this problem, we are given information about $f(0)$, $f'(0)$, and $f''(0)$, which strongly suggests that the center of the Taylor expansion is $c = 0$. Let's verify this. The Taylor series around $x = 0$ (Maclaurin series) is $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \ldots$. We are given: $f(0) = 2$ $f'(0) = 3$ $f''(0) = 12$ Substituting these values into the Maclaurin series form.

27 [Foundational] • Target: Term-by-term operations preserve the radius of convergence

Consider the Taylor series expansion of the function $f(x) = e^{kx}$ centered at $x = 0$. If the coefficient of the x^3 term in this expansion is $\frac{4}{3}$, what is the value of the constant k?

a. $k = 1$

b. $k = 2$ ✓

c. $k = 3$

d. $k = 4$

The Maclaurin series for e^u is given by $\sum_{n=0}^{\infty} \frac{u^n}{n!} = 1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \ldots$. For the function $f(x) = e^{kx}$, we substitute $u = kx$ into the Maclaurin series: $f(x) = e^{kx} = 1 + (kx) + \frac{(kx)^2}{2!} + \frac{(kx)^3}{3!} + \ldots$ $f(x) = 1 + kx + \frac{k^2x^2}{2} + \frac{k^3x^3}{6} + \ldots$ The coefficient of the x^3 term is $\frac{k^3}{6}$. We are given that the coefficient of the x^3 term is $\frac{4}{3}$. Therefore, we can set up the equation.

28 [Foundational] • Target: Building a new series by combining known ones

Consider the function $f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$. What is the value of the coefficient of the x^3 term in the Maclaurin series for $f(x)$?

a. $\frac{1}{3!}$

b. $\frac{1}{3}$

c. $\frac{1}{6}$ ✓

d. 1

The Maclaurin series for a function $f(x)$ is given by $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$. The given function is $f(x) = e^x$, which has the known Maclaurin series $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$. The coefficient of the x^3 term is $\frac{1}{3!}$. Since $3! = 3 \times 2 \times 1 = 6$, the coefficient is $\frac{1}{6}$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Ratio Test with factorial and power combinations**

In the Maclaurin series of $\sin(6x)$, what is the coefficient of the x^2 term (even power)?

a. 18

b. -18

c. 1

d. **0 ✓**

$\sin(6x)$ is an odd function, so its Maclaurin series has only odd powers of x; the coefficient of every even power (including x^2) is 0.

30 **[Intermediate]** • **Target: Using a power series to approximate a definite integral**

Consider the power series $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2 3^n}$. This series defines a function $f(x)$ over its interval of convergence. Let I be the interval of convergence for this power series. Determine which of the following statements is TRUE about $f(x)$ and its behavior at the endpoints of I.

a. **The radius of convergence is 3, and the series converges at both endpoints of the interval of convergence. ✓**

b. The radius of convergence is 3, and the series diverges at the left endpoint and converges at the right endpoint.

c. The radius of convergence is $1/3$, and the series converges at both endpoints of the interval of convergence.

d. The radius of convergence is $1/3$, and the series diverges at both endpoints of the interval of convergence.

We use the Ratio Test to find the radius of convergence. Let $a_n = \frac{(x-2)^n}{n^2 3^n}$. Then, $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(x-2)^{n+1}}{(n+1)^2 3^{n+1}} \cdot \frac{n^2 3^n}{(x-2)^n}\right|$ $L = \lim_{n\to\infty} \left|\frac{x-2}{3} \cdot \frac{n^2}{(n+1)^2}\right| = \left|\frac{x-2}{3}\right| \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^2 = \left|\frac{x-2}{3}\right| \cdot 1$ For convergence, we need $L1$.

31 **[Intermediate]** • **Target: Finding the radius of convergence**

Term-by-term differentiation is valid inside the radius of convergence. The derivative of $\sum_{n=0}^{\infty} x^n/(n+1)$ is:

a. $\sum x^n/n$

b. $\sum n x^{n-1}/(n+1)$

c. $1/(1-x)$ ✓

d. $\ln(1-x)$

Recognise $\int \frac{1}{1-x} dx = -\ln(1-x) = \sum x^n/(n+1)$; differentiate $\to 1/(1-x)$.

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32 [Intermediate] • Target: Finding the interval of convergence (testing endpoints)

Consider the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$. If the radius of convergence is $R=3$ and the center of the series is $a=2$, which of the following intervals represents the values of x for which the series converges?

a. $(-1,5)$ ✓

b. $[-1,5]$

c. $(-1,5]$

d. $[-1,5)$

The radius of convergence R of a power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ indicates that the series converges absolutely for $|x-a|R$. The center of the series is a. In this problem, we are given $R=3$ and $a=2$. Thus, the series converges for $|x-2|<3$. This inequality can be rewritten as $-3<x-2<3$. Adding 2 to all parts of the inequality, we get $-3+2<x<3+2$, which simplifies to $-1<x<5$. This interval is written in open interval notation as $(-1,5)$. The convergence at the endpoints.

33 [Intermediate] • Target: Representing functions as power series

Consider the function $f(x)=\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{(2n)!}$. Which of the following is the Maclaurin series for $\cos(x)$?

a. $\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{(2n)!}$ ✓

b. $\sum_{n=0}^{\infty}\frac{x^n}{n!}$

c. $\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)!}$

d. $\sum_{n=0}^{\infty}\frac{(-1)^n x^n}{n!}$

The Maclaurin series for $f(x)$ is a Taylor series expansion around $x=0$. We are given the series $f(x)=\sum_{0}^{\infty}\frac{(-1)^n x^{2n}}{(2n)!}$. Expanding this series, we get: For $n=0$: $\frac{(-1)^0 x^{2(0)}}{(2(0))!}=\frac{1\cdot x^0}{0!}=\frac{1}{1}=1$ For $n=1$: $\frac{(-1)^1 x^{2(1)}}{(2(1))!}=\frac{-1\cdot x^2}{2!}=-\frac{x^2}{2}$ For $n=2$: $\frac{(-1)^2 x^{2(2)}}{(2(2))!}=\frac{1\cdot x^4}{4!}=\frac{x^4}{24}$ So, $f(x)=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\cdots$. This is th.

34 [Intermediate] • Target: Differentiating and integrating power series

What is the Maclaurin series expansion for $f(x)=e^{3x}$?

a. $\sum_{n=1}^{\infty}\frac{(3x)^n}{n}$

b. $\sum_{n=0}^{\infty}(-1)^n\frac{(3x)^n}{n!}$

c. $\sum_{n=0}^{\infty}\frac{3x^n}{n!}$

d. $\sum_{n=0}^{\infty}\frac{(3x)^n}{n!}$ ✓

Substitute $u=3x$ into $e^u=\sum_{n=0}^{\infty}\frac{u^n}{n!}$ to obtain $\sum_{n=0}^{\infty}\frac{(3x)^n}{n!}$.

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35 [Intermediate] • Target: More radius/interval of convergence practice

Find the coefficient of x^4 in the Maclaurin series of e^{2x}.

a. $\frac{2}{3}$ ✓

b. $\frac{4}{3}$

c. $\frac{8}{3}$

d. $\frac{1}{12}$

$e^{2x} = \sum_{n=0}^{\infty} \frac{(2x)^n}{n!}$. Coefficient of x^4 is $\frac{2^4}{4!} = \frac{2}{3}$.

36 [Intermediate] • Target: Visualizing a power series approximation

Given $f^{(3)}(a) = 4$, find the coefficient of $(x-a)^3$ in the Taylor series of f centred at a.

a. $\frac{4}{3}$

b. $\frac{2}{3}$ ✓

c. 2

d. 4

Taylor coefficient formula: $\frac{f^{(3)}(a)}{3!} = \frac{4}{3!} = \frac{2}{3}$.

37 [Intermediate] • Target: Term-by-term operations preserve the radius of convergence

Maclaurin series for $\sin x$ is:

a. $\sum \frac{x^{2n}}{(2n)!}$

b. $\sum (-1)^n \frac{x^{2n+1}}{(2n+1)!}$ ✓

c. $\sum (-1)^n \frac{x^{2n}}{(2n)!}$

d. $\sum \frac{x^n}{n!}$

$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \ldots$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Building a new series by combining known ones • Cross-Lesson: 4.02 Area in Polar Coordinates

Taylor series for e^x at $x = 0$ is:

a. $\sum \frac{x^n}{n}$

b. $\sum \frac{x^n}{n!}$ ✓

c. $\sum x^n$

d. $\sum (-1)^n \frac{x^n}{n!}$

$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

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39 [Challenging] • Target: Ratio Test with factorial and power combinations • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Composite series: $e^{\sin x}$ up to x^3. Using $\sin x = x - x^3/6 + \cdots$ inside $e^u = 1 + u + u^2/2 + \cdots$:

a. $1 + x + \frac{x^2}{2}$ ✓

b. e

c. $1 + x + \frac{x^2}{2} + \frac{x^3}{6}$

d. $1 + x + \frac{x^2}{2} - \frac{x^3}{3}$

Substituting $u = x - \frac{x^3}{6} + \ldots$ into $e^u = 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + \ldots$ gives $1 + \left(x - \frac{x^3}{6}\right) + \frac{x^2}{2} + \frac{x^3}{6} + O(x^4) = 1 + x + \frac{x^2}{2}$.

40 [Challenging] • Target: Using a power series to approximate a definite integral • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

For a power series $\sum c_n x^n$, the radius of convergence R satisfies $1/R =$

a. $\limsup|c_n|^{1/n}$ ✓

b. $\limsup|c_{n+1}/c_n|$

c. $\sup|c_n|$

d. $|c_0|$

Applying Convergence Tests and Power Series principles, Cauchy-Hadamard formula. Therefore, the verified correct option is $\limsup|c_n|^{1/n}$.

41 [Challenging] • Target: Finding the radius of convergence • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Using the power series $1/(1-x) = \sum x^n$, find $\sum n x^{n-1} =$

a. $1/(1-x)$

b. $1/(1-x)^2$ ✓

c. $x/(1-x)^2$

d. $1/(1+x)^2$

Applying Convergence Tests and Power Series principles, Differentiate geometric series. Therefore, the verified correct option is $1/(1-x)^2$.

42 [Challenging] • Target: Finding the interval of convergence (testing endpoints) • Cross-Lesson: 1.01 Integration by Parts

Radius of convergence of $\sum x^n/n$ is $R =$

a. 0

b. e

c. ∞

d. 1 ✓

Applying Convergence Tests and Power Series principles, $R = \lim n/(n+1) = 1$. Therefore, the verified correct option is 1.

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43 [Challenging] • Target: Representing functions as power series • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

A power series $\sum c_n x^n$ with radius of convergence R converges absolutely for:

a. no x

b. all x

c. **|x|<R ✓**

d. $|x| > R$

By definition of the radius of convergence, the series converges absolutely inside $(-R, R)$ and diverges outside $[-R, R]$.

44 [Challenging] • Target: Differentiating and integrating power series • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Find radius of convergence: $\sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2} x^n$:

a. $1/4$ ✓

b. $1/2$

c. ∞

d. 4

Central binomial: $\binom{2n}{n} x^n$; $|c_n|^{1/n} \to 4 \Rightarrow R = 1/4$.

45 [Challenging] • Target: More radius/interval of convergence practice • Cross-Lesson: 3.01 Parametric Equations

If the radius of convergence of $\sum a_n x^n$ is R, then the radius of $\sum a_n x^{2n}$:

a. R

b. R^2

c. $\sqrt{R}$ ✓

d. $2R$

Substitute $y = x^2$: $|y| < R \Rightarrow |x|^2 < R \Rightarrow |x| < \sqrt{R}$.

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